



江苏省苏州中学国际部
Suzhou High School – International Division

SZZX-ID Bridge Program Physics

Apparent weight

Class Outline

Title of Unit	Forces and Newton's Laws
Title of Lesson	Apparent weight
Starter Activity	Revision Newton's laws
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	Lecture: Explain apparent weight in elevators with examples, Explain the definition of overweight Demonstration: The mechanism of parabolic flight Inquiry: explain why we observed overweight and underweight on a scale. overweight and underweight in life – homework
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	Homework instruction
	International Mindedness:
	International space station
Lesson Objectives	By the end of the lesson, students will be able to: - Understand what determines the apparent weight (the weight we feel, measure and see) - Apply the knowledge of Newton's second law to determine apparent weight when knowing acceleration, or vice versa - Understand and explain weightlessness and overweight
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	Communication: Clearly communicating complex ideas in response to open-ended questions : example of overweight or underweight in life
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	Communicator, inquirer
Teaching Activities	Class discussion: how to define weightlessness? (taking learning from the definition of overweight) Video: parabolic flight Class group discussion: how to create overweight and weightlessness on a scale
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	Explain homework - describe an overweight or underweight case in life

Vocabulary



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Apparent 明显的/表面看来

Perceive 感知/注意

Overweight 超重

Weightless 失重

Altitude 海拔

Radius 半径

Scale 尺度/称

Astronaut 宇航员

Weight



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Weight is the force resulted from gravitational attraction between the mass m of a body and the mass of the planet on which the body is placed.

$$W = mg$$

where m is the mass of the body and g is the acceleration due to gravity
Unit of g : N kg^{-1} or m s^{-2}

Measuring weight



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By measuring normal force on a bathroom scale



By measuring tension in a spring scale

When we measure our mass by a scale, what are we actually measuring?

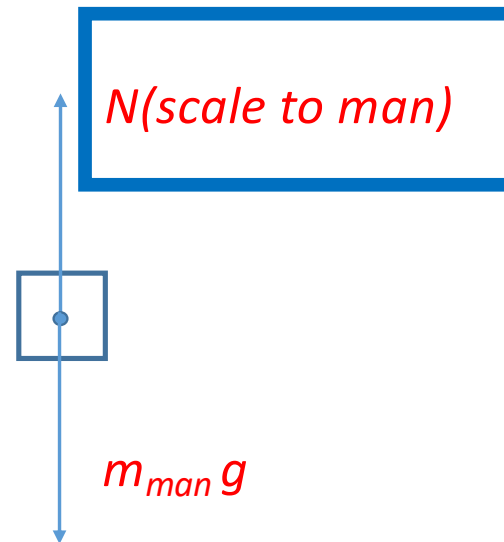
Measuring weight



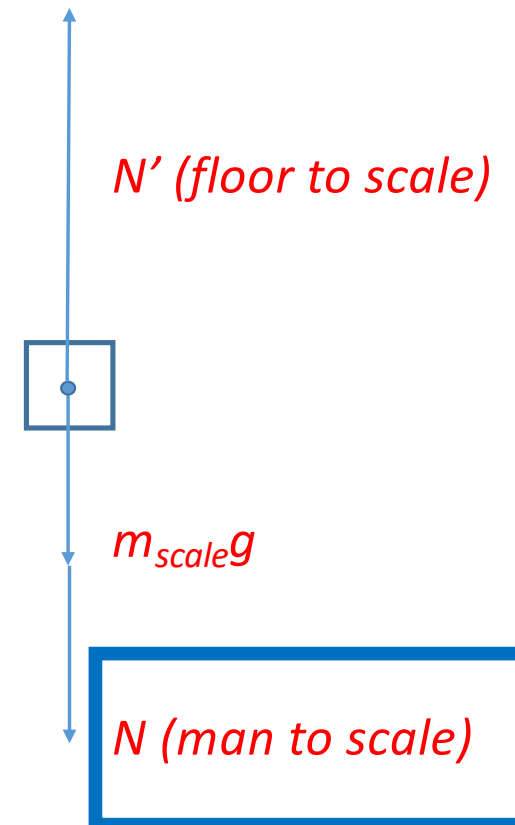
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The man



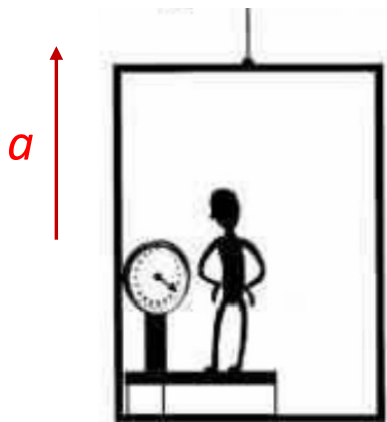
The scale



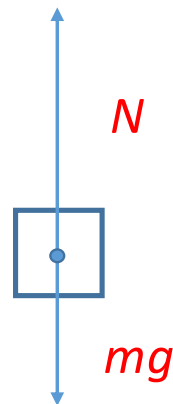
When you stand on a scale, which force determines the reading from the scale?



Measuring weight in an elevator



When the elevator is accelerating upwards:



Let N be the normal force acting on the man

$$F_{net} = N - mg = ma$$

$$N = m(g + a)$$

The normal force acting on the scale equals to the normal force acting on the man (Newton's third law)

$$N_{scale} = m(g + a)$$

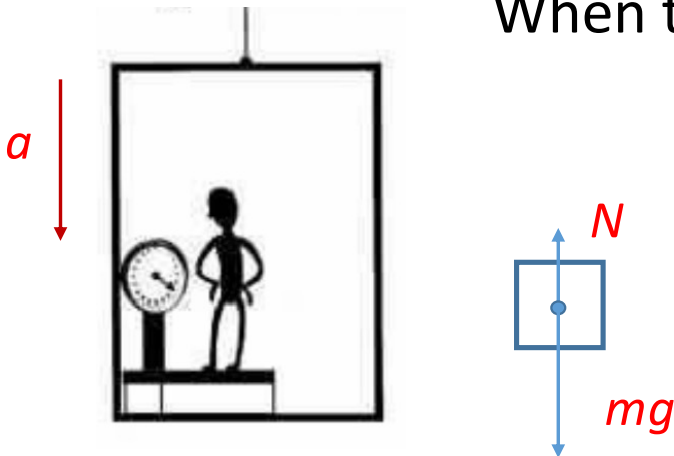
What weight will be perceived by the man? How does the apparent weight change with the *acceleration*?

Measuring weight in an elevator



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When the elevator is accelerating upwards:



Let N be the normal force acting on the man

$$F_{net} = mg - N = ma$$

$$N = m(g - a)$$

The normal force acting on the scale equals to the normal force acting on the man (Newton's third law)

$$N_{scale} = m(g - a)$$

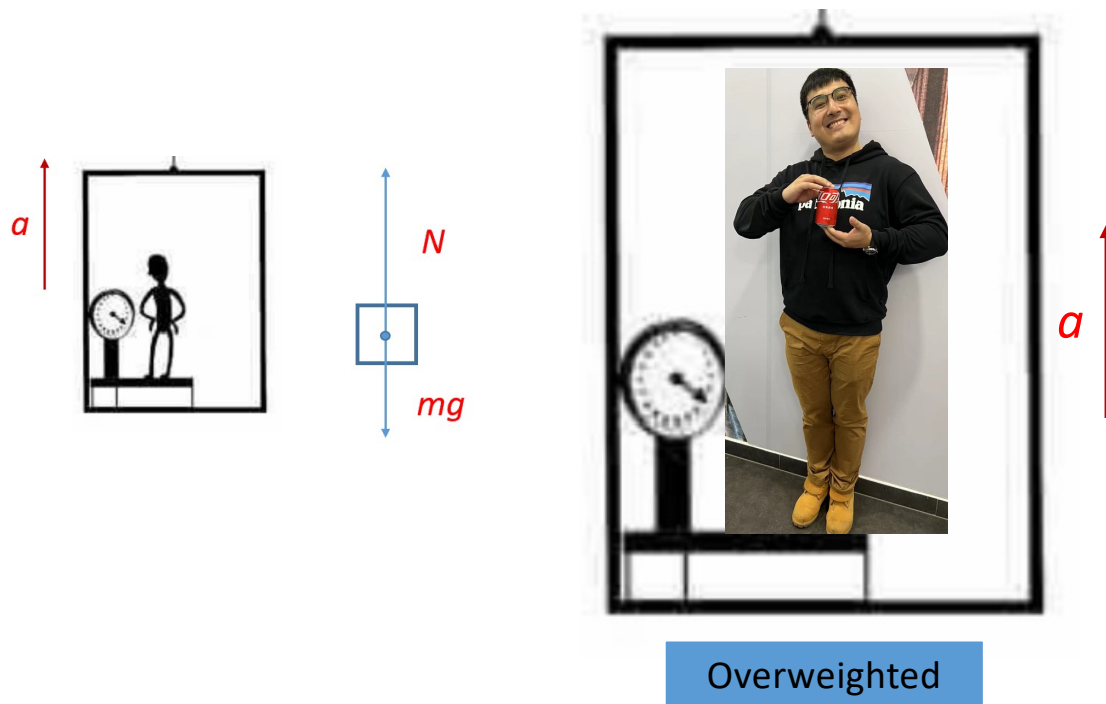
- What weight will be perceived by the man?
- How does the **apparent weight** change with a downwards?
- What happens when a reaches g ?
- Would the magnitude of a exceed g ?

Overweight and weightlessness

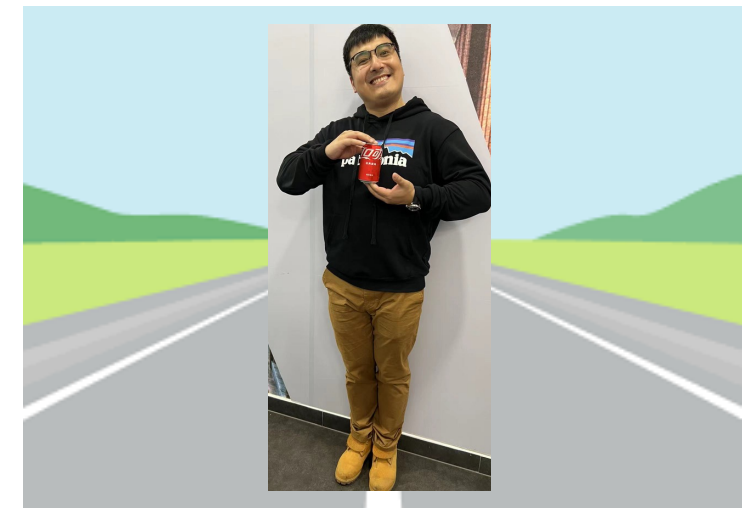


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Overweight: when the normal force (or tension) is more than the actual weight (mg), or, when the **apparent weight** is larger than the actual weight.



equilibrium



Still overweight!
(but in a different way)

Overweight and weightlessness



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“-less” means lack of something, or without something

- Motionless: lack of motion, or stationary
- Speechless: lack of word
- Effortless: without effort

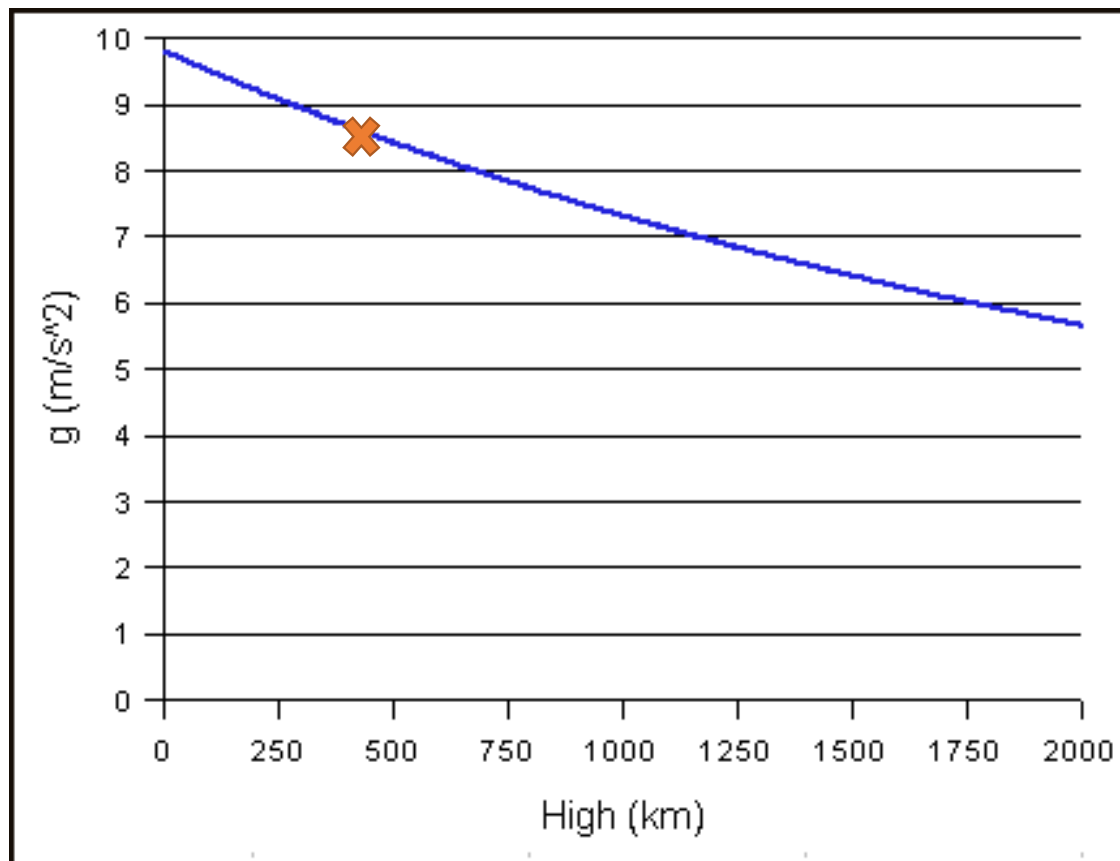
How do you interpret “weightless”?

- Lack of the sensation of weight (or apparent weight)
- Lack of the normal force or tension that would have been resulted from weight in other situations
- Lack of acting forces other than force of gravity

Weightlessness



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Height of the China Space Station: 380km
(~6% of the earth's radius)
Force of gravity is still quite strong!
Yet, astronauts will experience weightlessness.

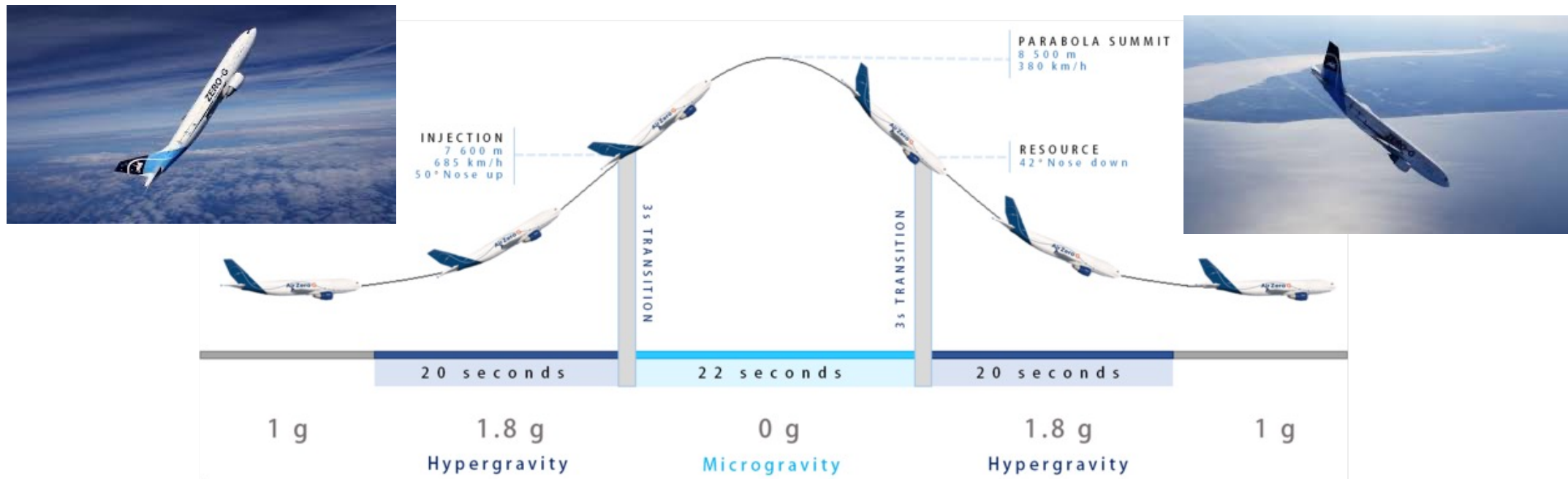


How can we create overweight and weightlessness?



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Parabolic flight: experience overweight, weightlessness, Moon gravity and mars gravity in the same flight



Picture from: The European Space Agency, https://www.esa.int/Education/Fly_Your_Thesis/Parabolic_manoeuvres

How can we create overweight and weightlessness?



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The option before we can afford a ride on the parabolic flight

Student activity:

Use a spring scale to create overweight and weightlessness.

- Stand up quickly and squat quickly, observe the readings
- Explain what happened in this process that causes the readings to change. Use a diagram to assist your explanation.
- what were the max. and min. readings (you can use an approximate)? Can you work out the max. acceleration in this process?

How can we create overweight and weightlessness?



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Homework:

- Observe an overweight or weightlessness case in life
- Describe the case with everyday words and also with physics terms/equations
- Use a free-body diagram to illustrate
- If you can, come out with some quantitative description (calculation with numbers) of the motions and forces.